Homa Harshitha Vuddanti

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.27
Nano Junior College, Nallakunta	Class XII (SSC)	2024	99.00%
Johnson Grammar School, Nacharam	Class X (CBSE)	2022	98.20%

SKILLS

- **Tools:** Cadence Virtuoso, LTspice, Latex, Quartus Prime, ModelSim, Arduino IDE
- **Programming Languages:** C, Python, Verilog HDL, MATLAB (Basics)

RELEVANT COURSES

Circuit Design: Circuits and network analysis, Analog Electronics, Semiconductor Device Fundamentals

Digital & Embedded Systems: Digital Systems, Digital Systems lab, Random Processes, Machine Learning

Programming: Introduction to programming (C), Data Structures and Algorithms (C), Scientific Programming

Mathematics: Probability and Random Processes, Differential Equations and Transform Techniques, Vector Calculus

Signal Processing: Signal Processing, Digital Signal Processing lab

Other core courses: Control Systems, Engineering Electromagnetics, Power System Analysis

RESEARCH EXPERIENCE

CMOS Image Sensor Research

- **Studied** CMOS image sensor architectures, including **3T/4T Active Pixel Sensors (APS), charge transfer mechanisms, correlated double sampling (CDS), and pixel readout techniques**, through standard reference literature to establish a foundation for analog image sensor design.
- Designed, simulated, and laid out a CMOS inverter in **Cadence Virtuoso** using an **industry-standard CMOS Process Design Kit (PDK)**, validating the design through **DRC** and **LVS** and gaining hands-on experience with the complete schematic-to-layout IC design flow.
- Currently developing simulation blocks for **correlated double sampling (CDS)** and **ramp generation** circuits in Cadence Virtuoso, extending the design flow toward CMOS image sensor analog front-end building blocks.

PROJECTS

Pipelined FIR Filter and FFT on FPGA

- Implemented a **100-tap FIR filter** in Verilog on an Intel FPGA using **direct, symmetric, and pipelined** architectures.
- Compared different FIR implementations by analyzing latency, resource utilization, and timing after synthesis in **Quartus Prime**.
- Verified the designs using ModelSim testbenches and fixed-point input data.
- Implemented FFT processing using the Intel **FFT IP core** and compared its functionality with the custom FIR implementations.

Sampling and Reconstruction System using an 8-bit SAR ADC

- Designed and implemented an **8-bit microcontroller-assisted SAR ADC**.

- Integrated a **sample-and-hold** stage, LM311 comparator, **R-2R ladder DAC**, and reconstruction low-pass filter.
- Demonstrated analog signal **sampling and reconstruction** using the complete SAR ADC system.
- Investigated **Nyquist sampling**, aliasing, and signal reconstruction by varying the effective sampling frequency through SAR conversion timing.
- Experimentally validated the Sampling Theorem using the developed hardware setup.
- Characterized converter performance by measuring **effective resolution**, **LSB voltage**, **offset error**, **gain error**, **DNL**, **INL**, and conversion time using oscilloscope measurements.

Analog Circuit Design and Simulation | LTspice

- Designed and simulated **RC Phase Shift** and **Wien Bridge oscillators** for ~50 kHz sinusoidal signal generation through RC network design.
- Implemented **op-amp based triangular and sawtooth waveform generators** operating at ~10 kHz and verified transient response in LTspice.
- Designed a **Schmitt trigger** with **±200 mV switching thresholds** by optimizing the positive feedback network and validating hysteresis characteristics.
- Analyzed the impact of **op-amp non-idealities** on oscillator frequency and waveform characteristics, refining component values to improve design accuracy.

POSITIONS OF RESPONSIBILITY

Sunshine Club - UG Mentor & Mentor Coordinator

- Mentored junior students and coordinated the undergraduate mentoring program.
- Received the **Star Performer Award** for outstanding contribution.

Media Council - Editorial Coordinator

- Created digital content and supported publicity for institute events.

Sunshine Club - UG Buddy

- Mentored first-year students; received the **Best Buddy Award**.

EXTRACURRICULAR

- Member of the institute **Volleyball** team under the **National Sports Organisation** (NSO).
- Represented IIT Hyderabad in Volleyball at **Inter IIT 2025.**